



NPTEL Course on “Computer System Optimization for ML”

Lecture #14



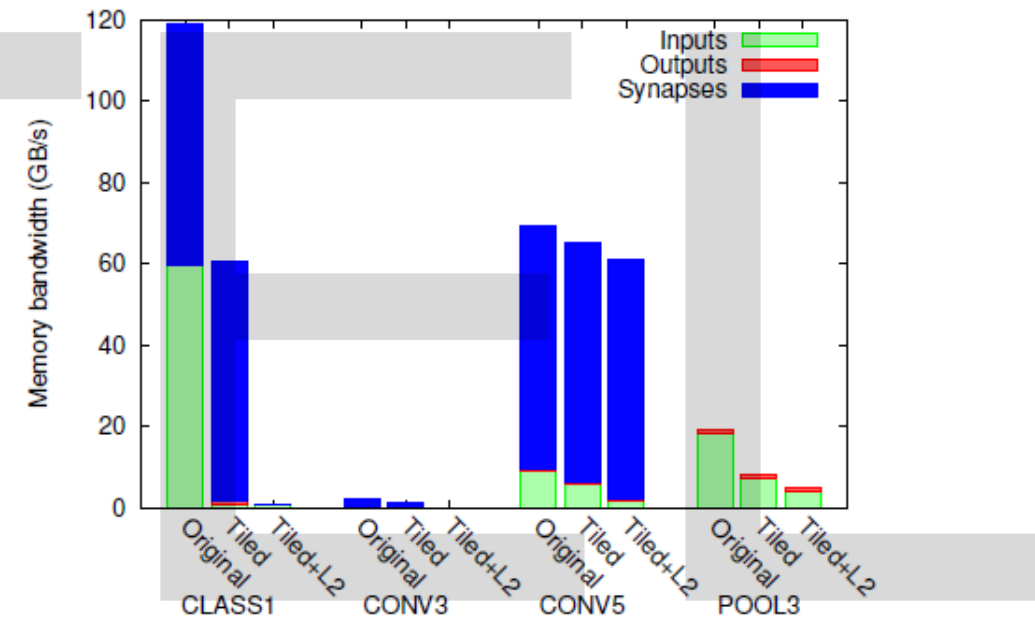
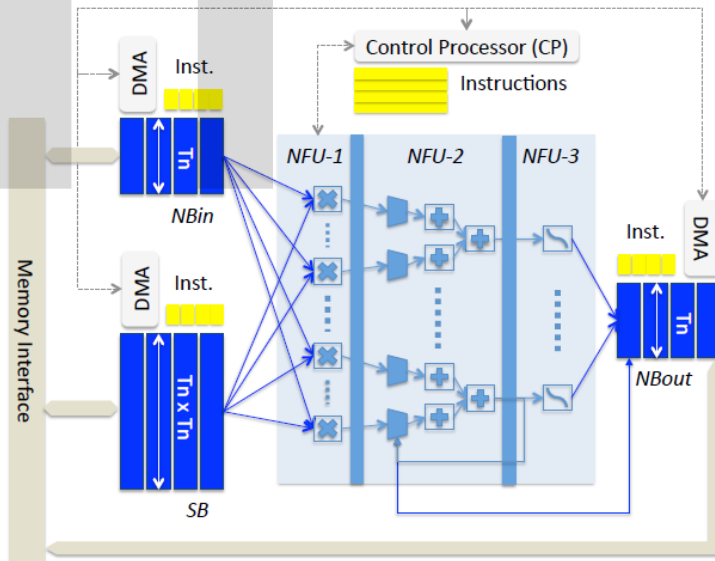
Agenda

- **Recap**
- **DNN Accelerator (contd)**

NPTREL

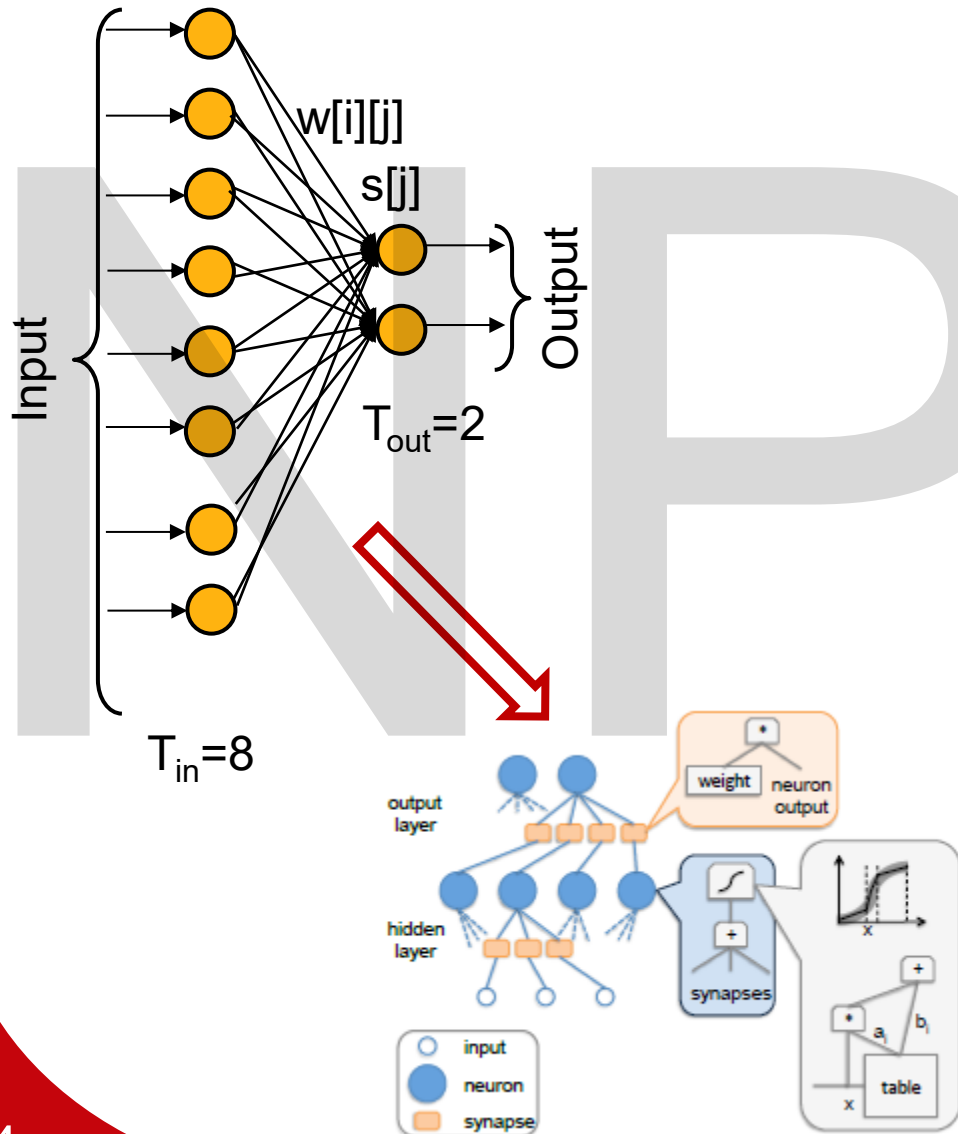
DianNao (Computer) [1]

- One of the earliest accelerator for DNN models
- Tiling to reduce memory usage
 - Classifier layer
 - Convolutional layer
 - Pooling layer
- An architecture tailored for DNN inference



[1] Chen, Tianshi, et al. "Diannao: A small-footprint high-throughput accelerator for ubiquitous machine-learning." ACM SIGARCH Computer Architecture News 42.1 (2014): 269-284.

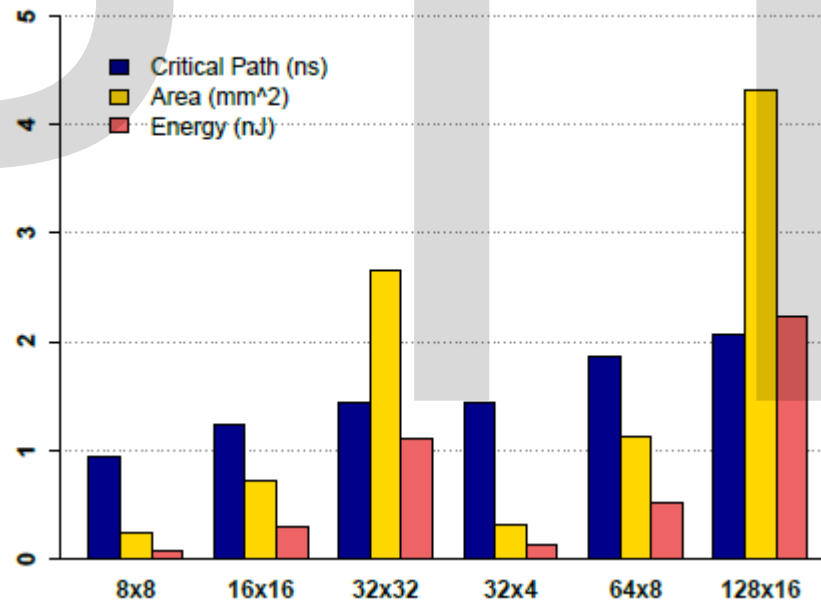
A Naïve Hardware Implementation



- **Fully layout the neurons and the synapses on the silicon**
 - Neurons: logic circuits performing computation
 - Synapses: latches/RAMs storing the weights
 - Non linear functions are implemented as piecewise linear function
- **Advantage**
 - High speed: distance travelled by the data is less
- **Disadvantage**
 - Area increases drastically with increase in the number of neurons

Performance of the Naïve Implementation

- Area and energy suffers a lot with increasing number of input and output neurons



Accelerator Architecture in DianNao

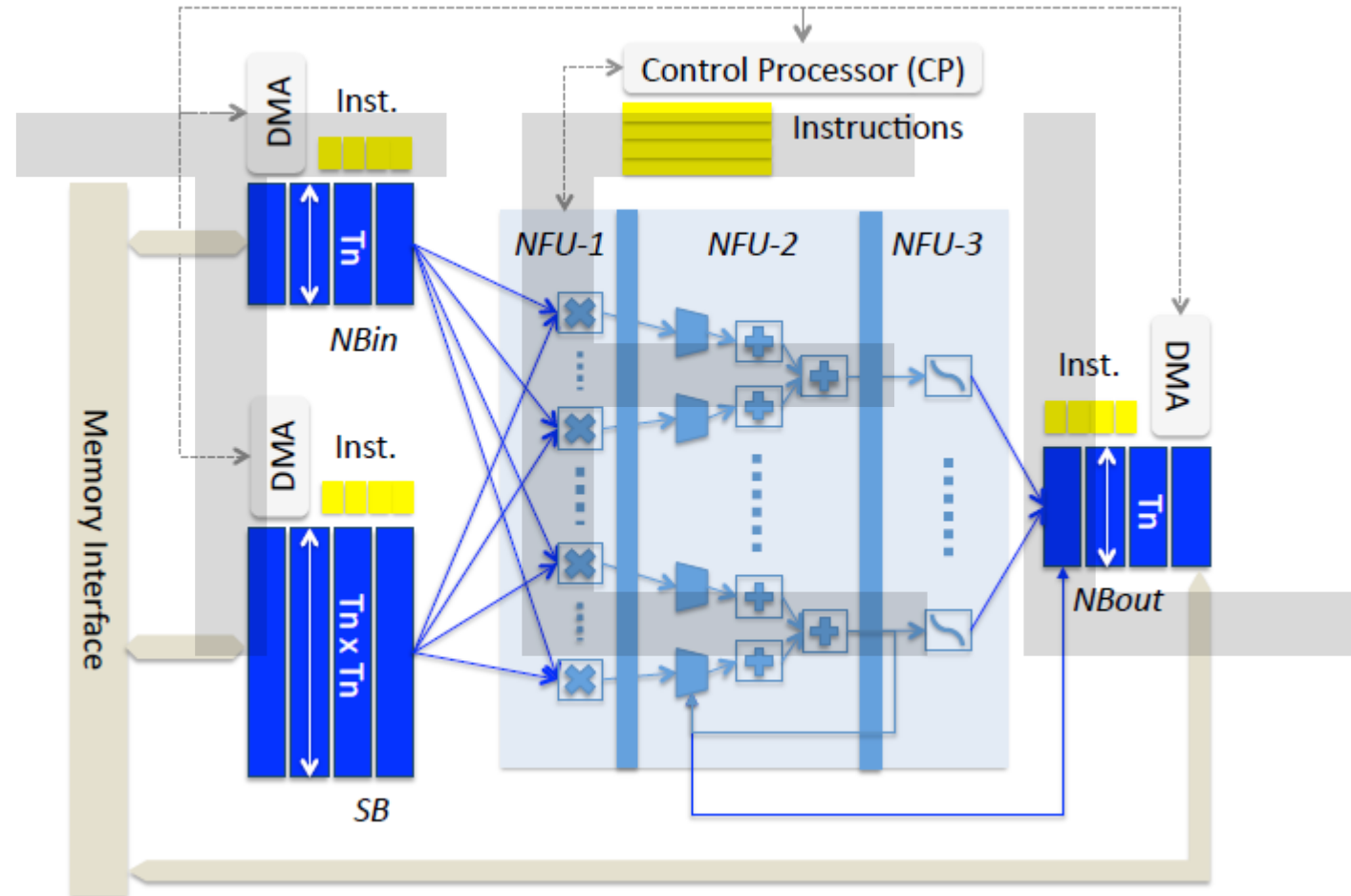
■ Storage

- NBin: To store input activations
- SB: To store synapses
- NBout: To store outputs

■ Neural functional unit (NFU)

- NFU-1: multiplier
- NFU-2: adder tree
- NFU-3: non linear

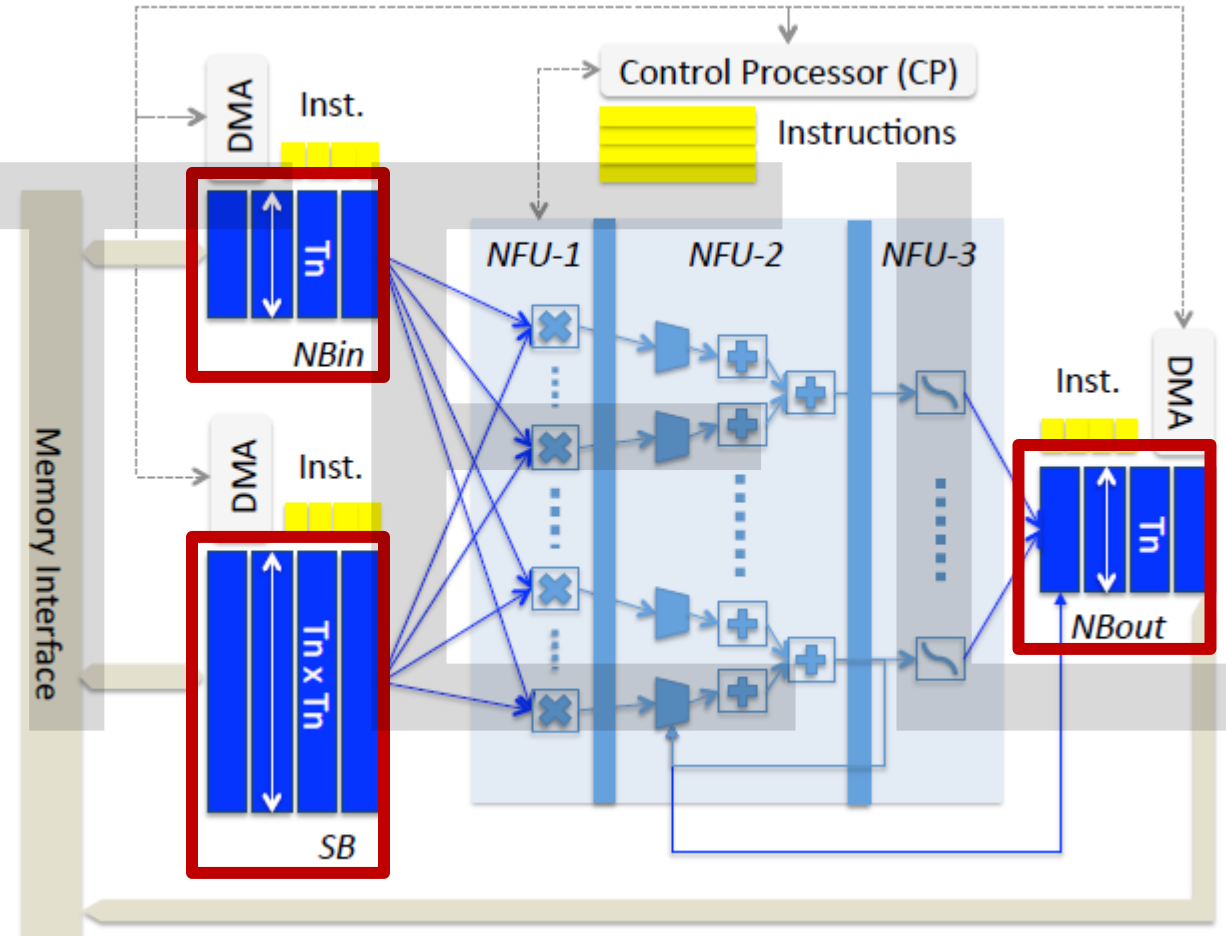
■ Control instructions



Storage (1)

- **Split buffers: Three different buffers instead of single cache**
 - Reduced conflict
 - Reduced access time
- **Scratchpad memory**
- **Point to be noted**
 - Different width of NBin/NBout and SB enables reading of different width of data

```
for (i=0; i<To; i++) {  
  for (j=0; j<Ti; j++) {  
    s[i] += a[j].w[i][j];  
  }  
}
```

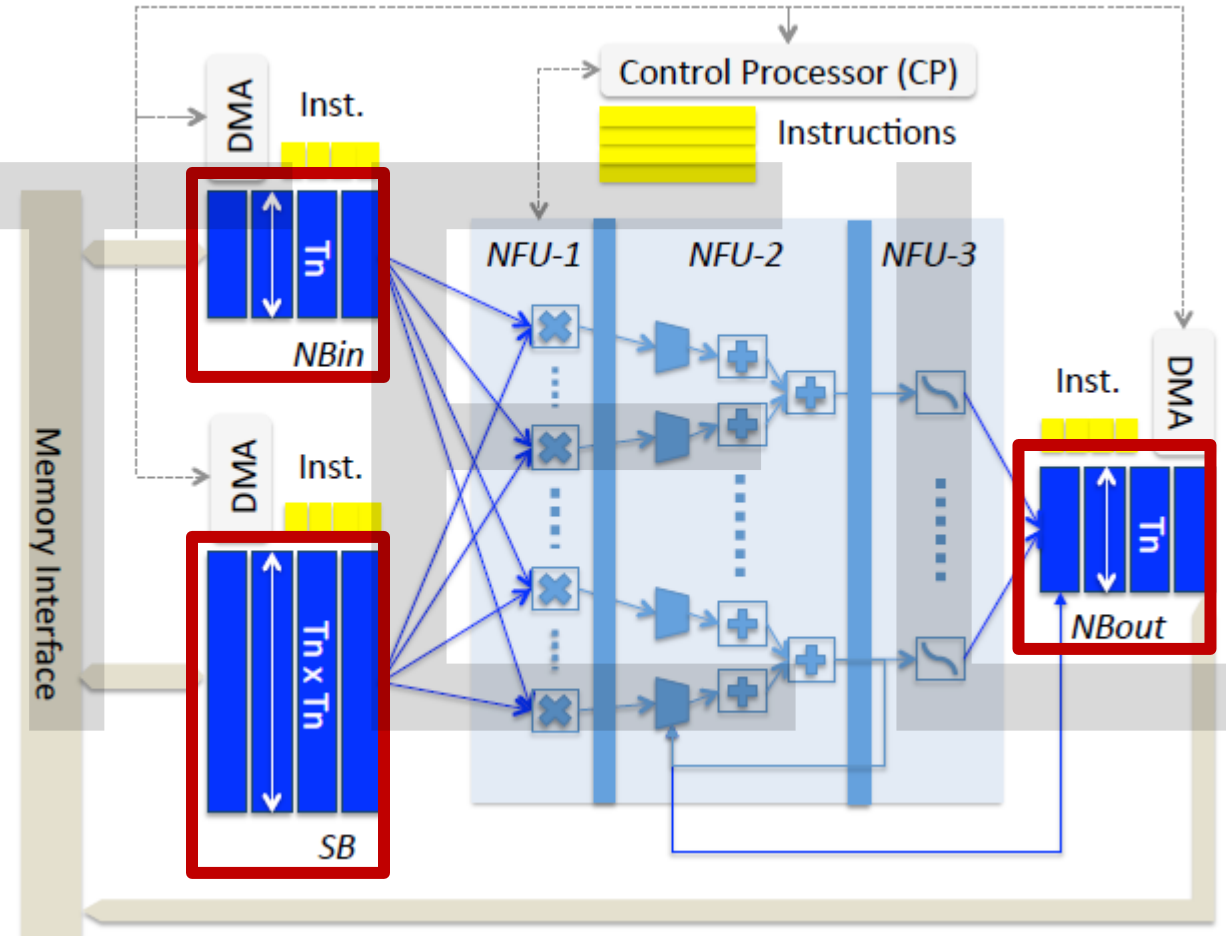


Storage (2)

- **NBout is a Circular buffer**

- Partial sums are pushed into NBout instead of sending them back into the main memory and fetching them again
- Output is read from NBout only when all the input neurons are integrated for the partial sum
- Useful while tiling

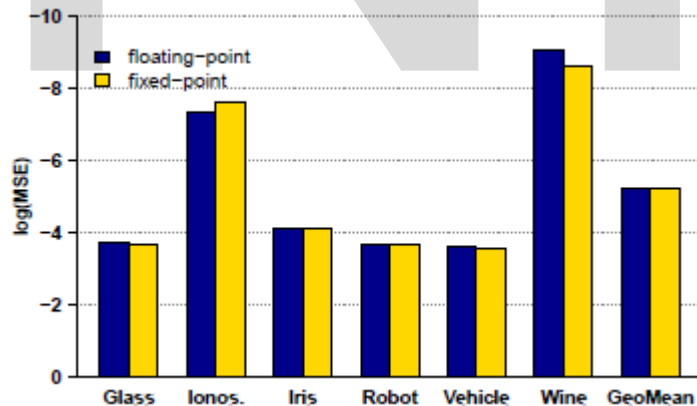
```
for (i=0; i<To; i++) {  
  for (j=0; j<Ti; j++) {  
    s[i] += a[j].w[i][j];  
  }  
}
```



NFU-1: Multipliers

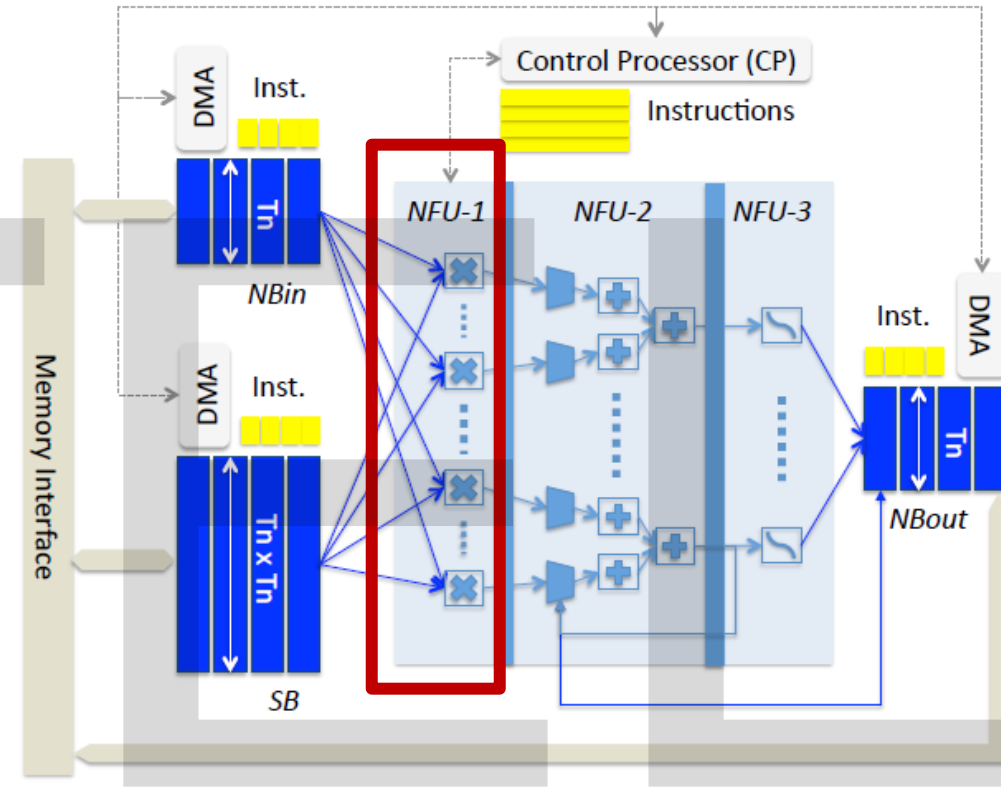
- Convolutional layers require multiplication
- 16-bit fixed point multipliers are used instead of 32-bit floating point multipliers

Error



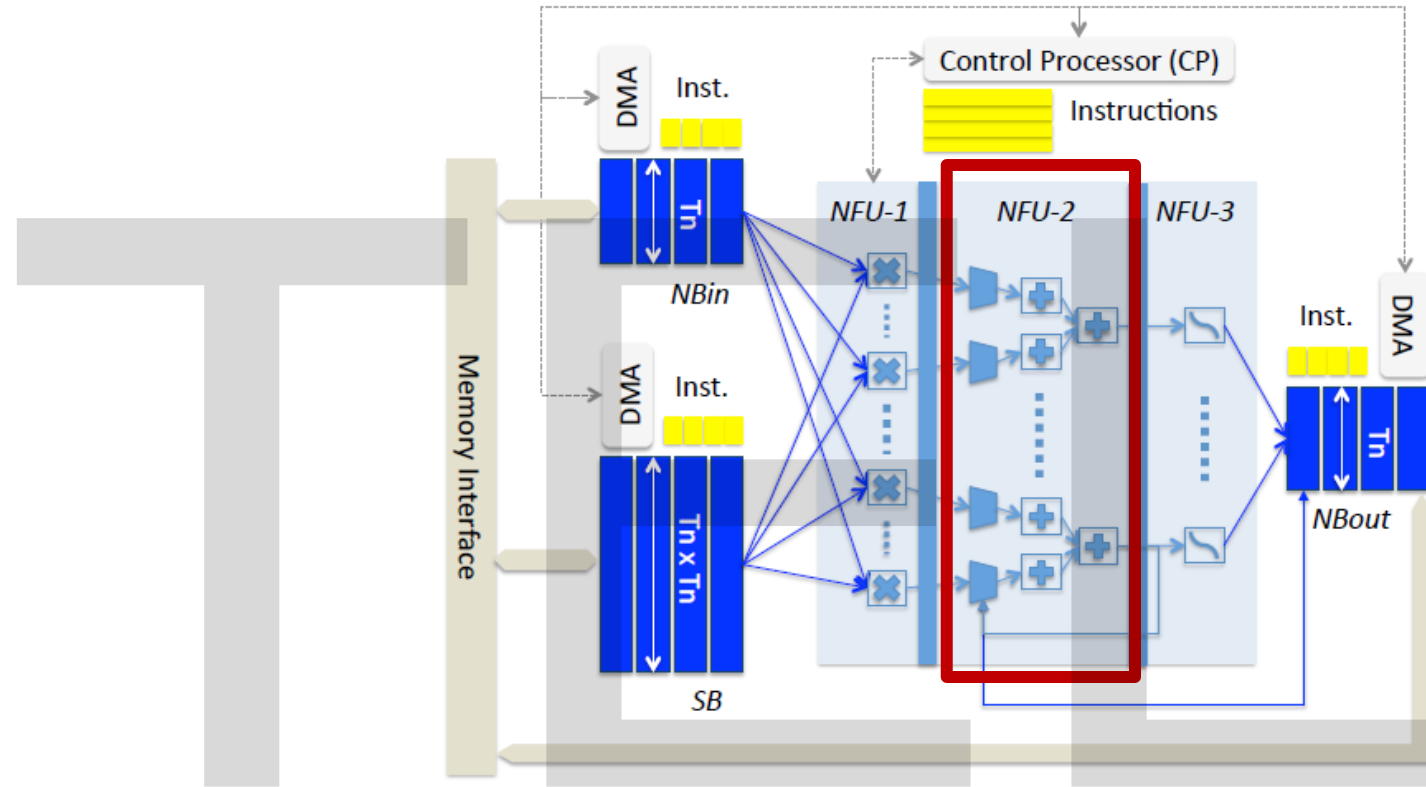
Performance

Type	Area (μm^2)	Power (μW)
16-bit truncated fixed-point multiplier	1309.32	576.90
32-bit floating-point multiplier	7997.76	4229.60



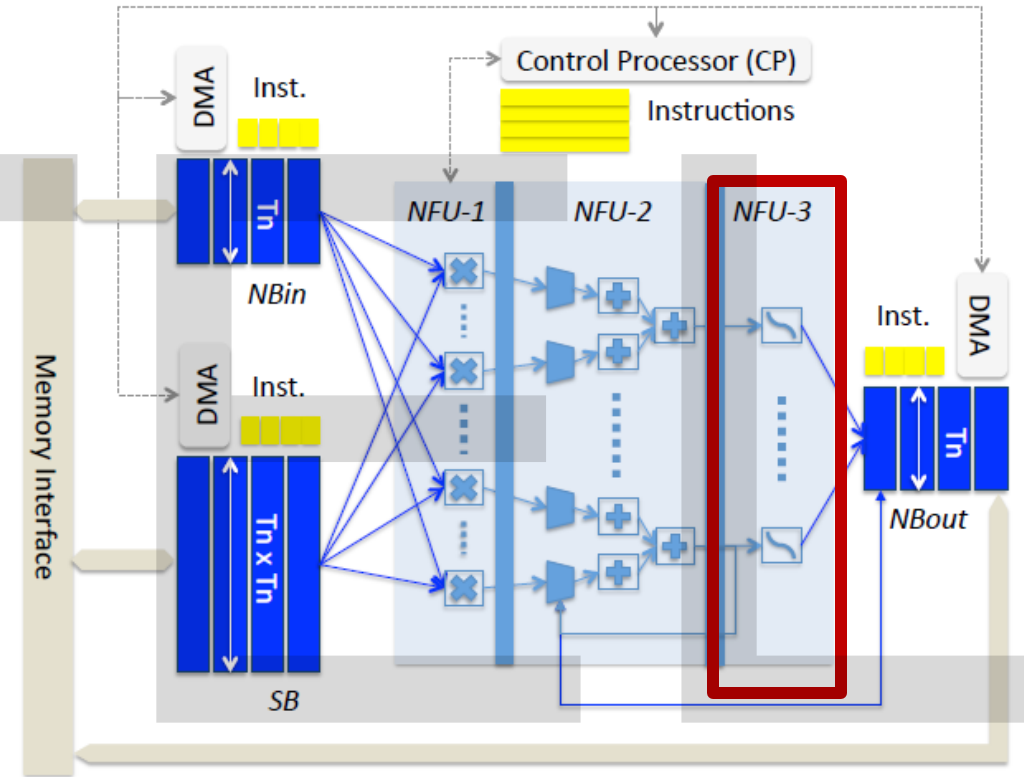
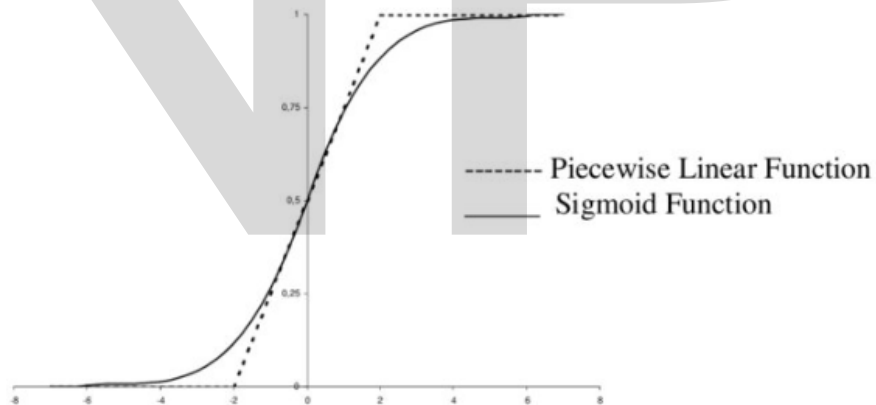
NFU-2: Adders

- Adder trees
- Advantages of adder trees
 - Less generate and propagate
 - Less latency



NFU-3: Non linear Unit

- **Involves non-linear operations**
 - Power of exponential, division etc.
- **Any optimization idea?**
 - Implement as piecewise linear function



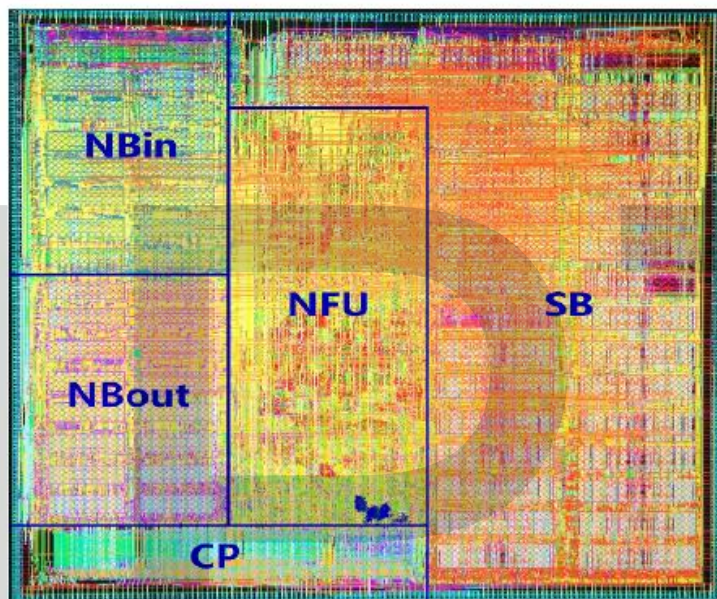
Control Instruction Format

An example:

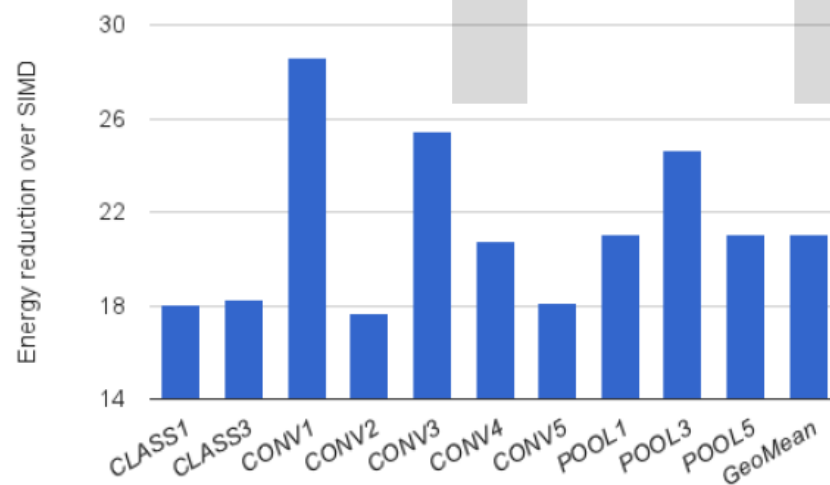
$T_{in}=8192, T_{out}=256$, 64-entry buffers

12

Results

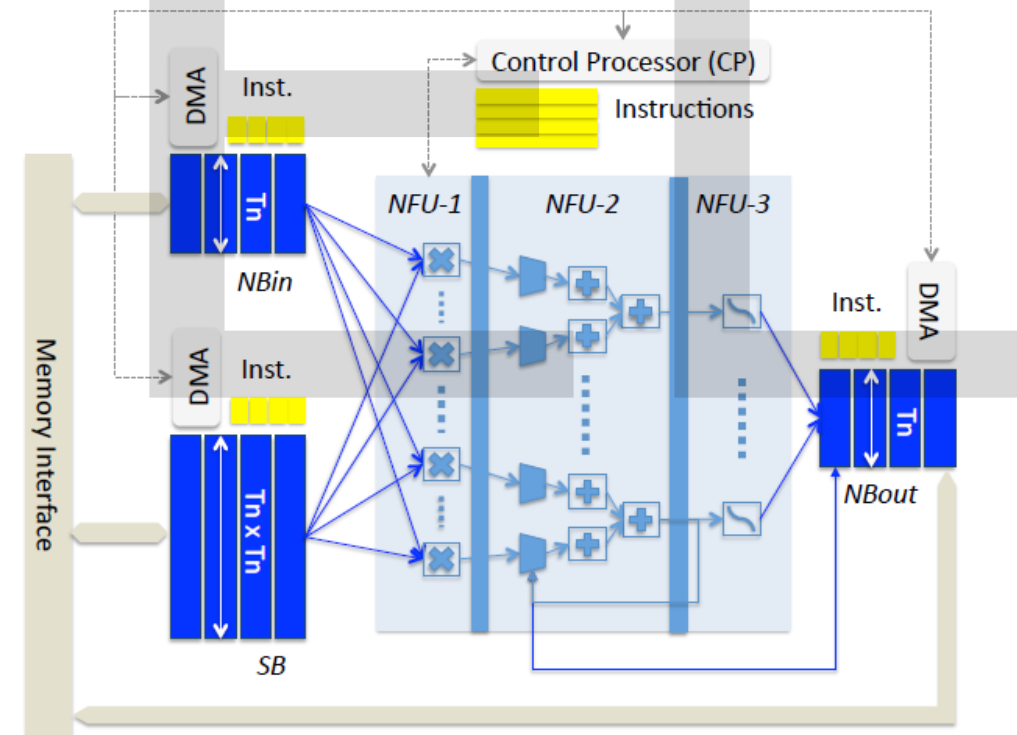


Component or Block	Area in μm^2	(%)	Power in mW	(%)	Critical path in ns
ACCELERATOR	3,023,077		485		1.02
Combinational	608,842	(20.14%)	89	(18.41%)	
Memory	1,158,000	(38.31%)	177	(36.59%)	
Registers	375,882	(12.43%)	86	(17.84%)	
Clock network	68,721	(2.27%)	132	(27.16%)	
Filler cell	811,632	(26.85%)			
SB	1,153,814	(38.17%)	105	(22.65%)	
NBin	427,992	(14.16%)	91	(19.76%)	
NBout	433,906	(14.35%)	92	(19.97%)	
NFU	846,563	(28.00%)	132	(27.22%)	
CP	141,809	(5.69%)	31	(6.39%)	
AXIMUX	9,767	(0.32%)	8	(2.65%)	
Other	9,226	(0.31%)	26	(5.36%)	



DaDianNao (A big computer) [2]

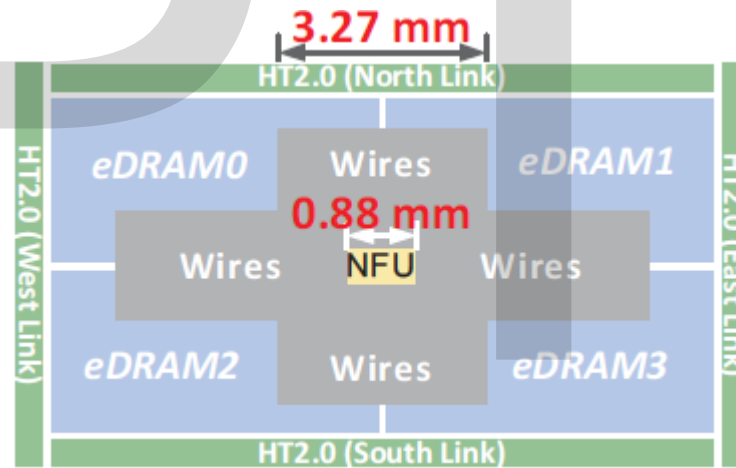
- **DianNao consists of a single NFU unit**
 - Not suitable for executing large DNNs
- **DaDianNao integrates many of DianNao-NFU on a single chip**
 - Synapses close to neuron
 - eDRAM instead of SRAM



[2] Chen, Yunji, et al. "Dadiannao: A machine-learning supercomputer." *2014 47th Annual IEEE/ACM International Symposium on Microarchitecture*. IEEE, 2014.

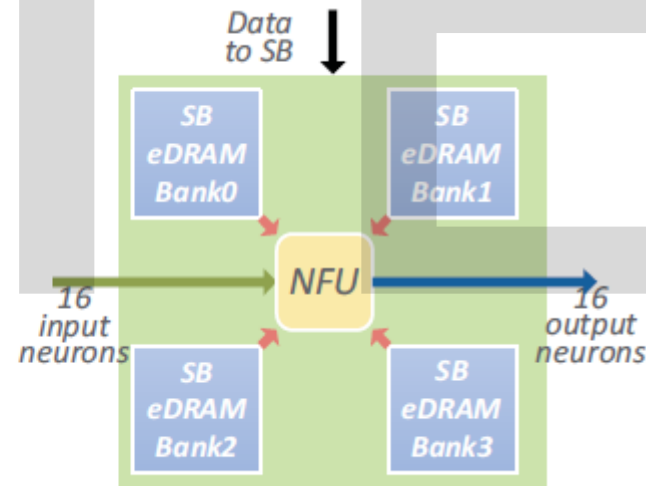
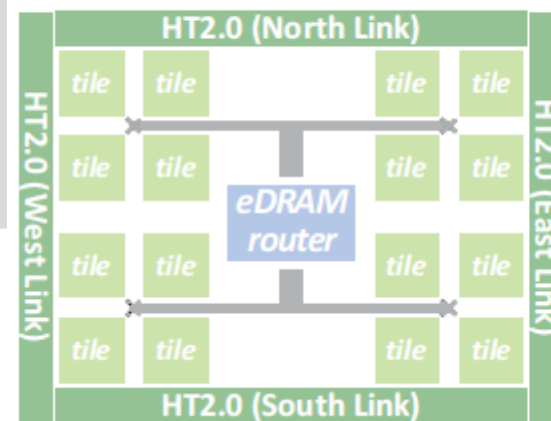
eDRAM placement

- **Each NFU is connected to four eDRAM banks**
 - Synapses are stored in the eDRAM
 - Synapse rows are interleaved

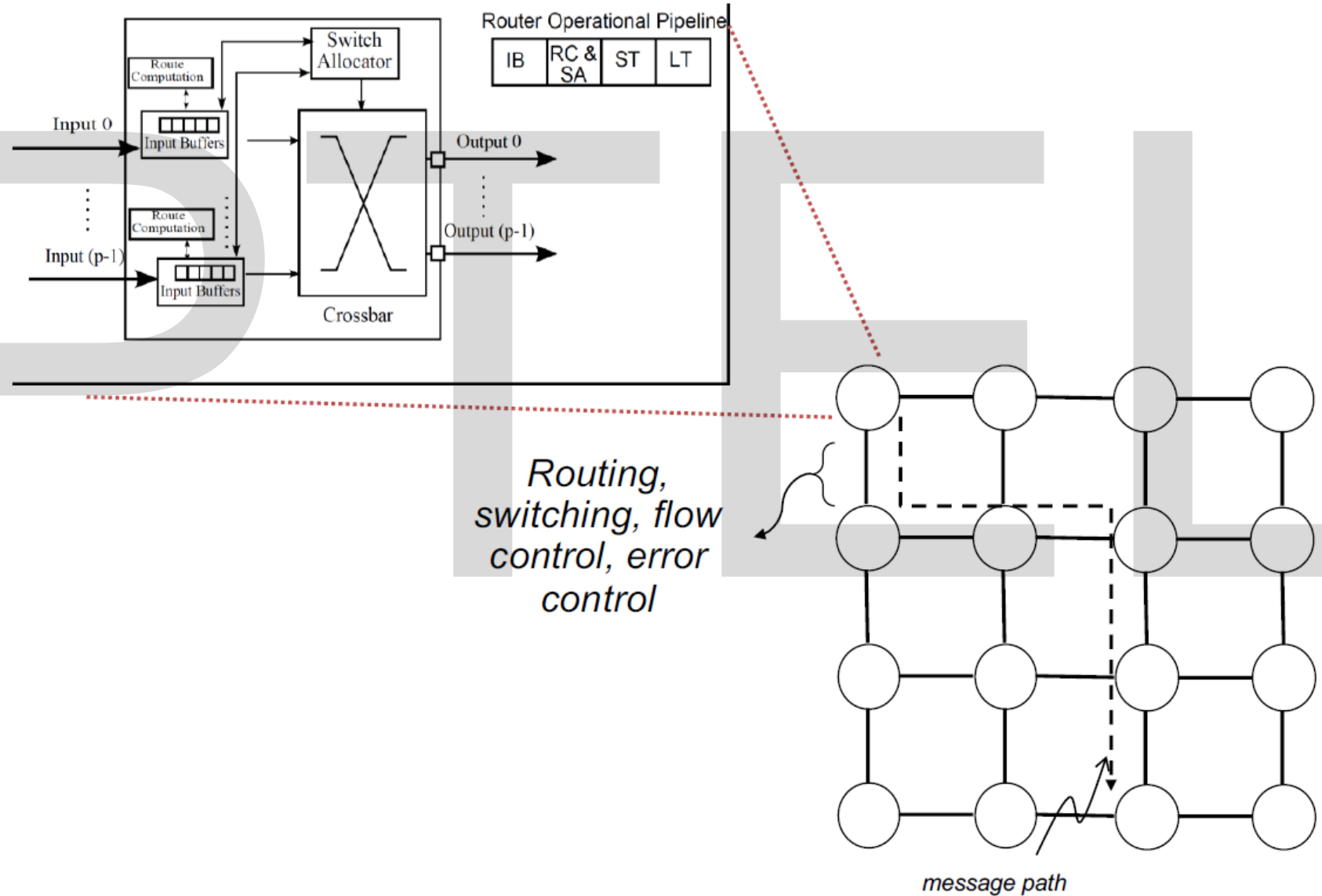


Tile-based Organization (1)

- Each tile consists of 4 eDRAM banks and an NFU (previous slide)
- 16 such tiles are integrated with
 - Two central eDRAM banks with fat-tree interconnect
 - Tiles are connected with mesh NoC having routers 5×5 crossbars (details in next slide)

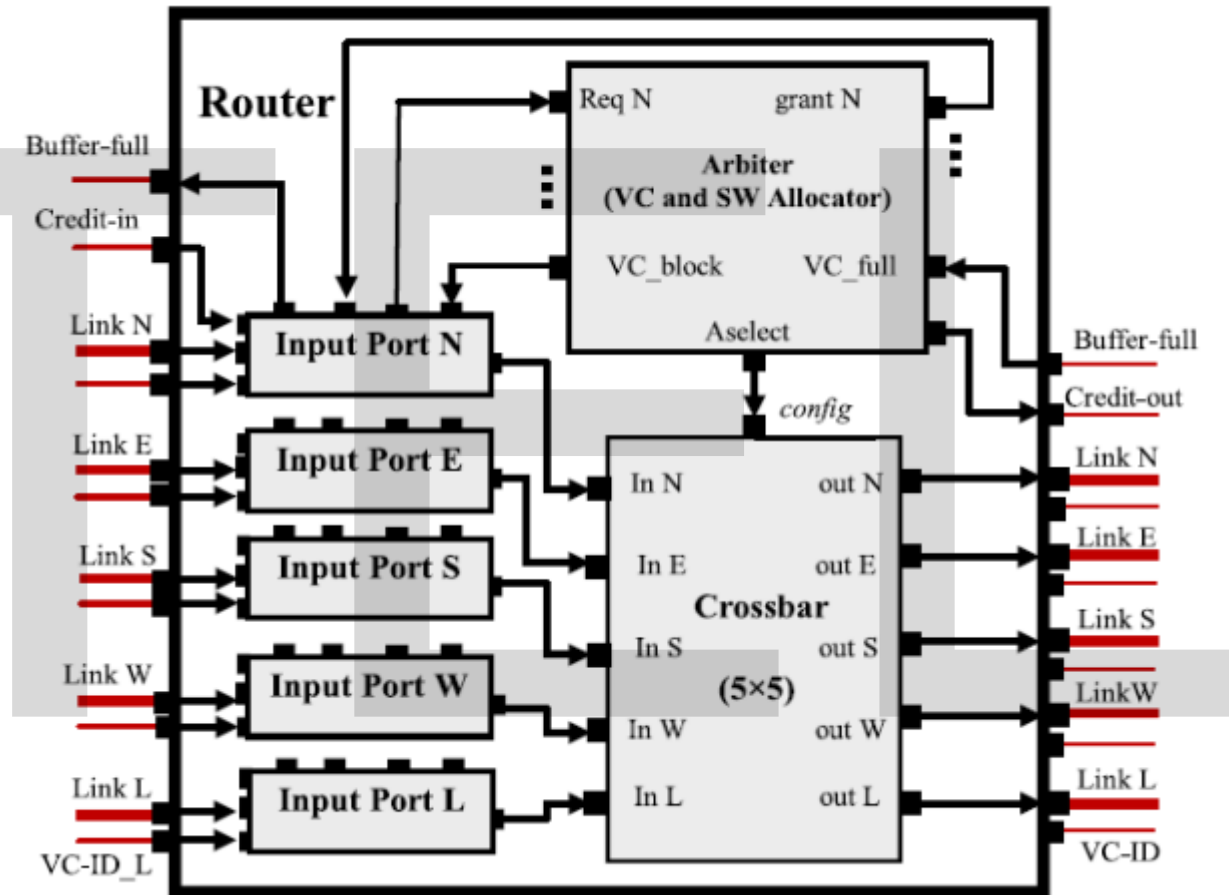


Mesh NoC Architecture

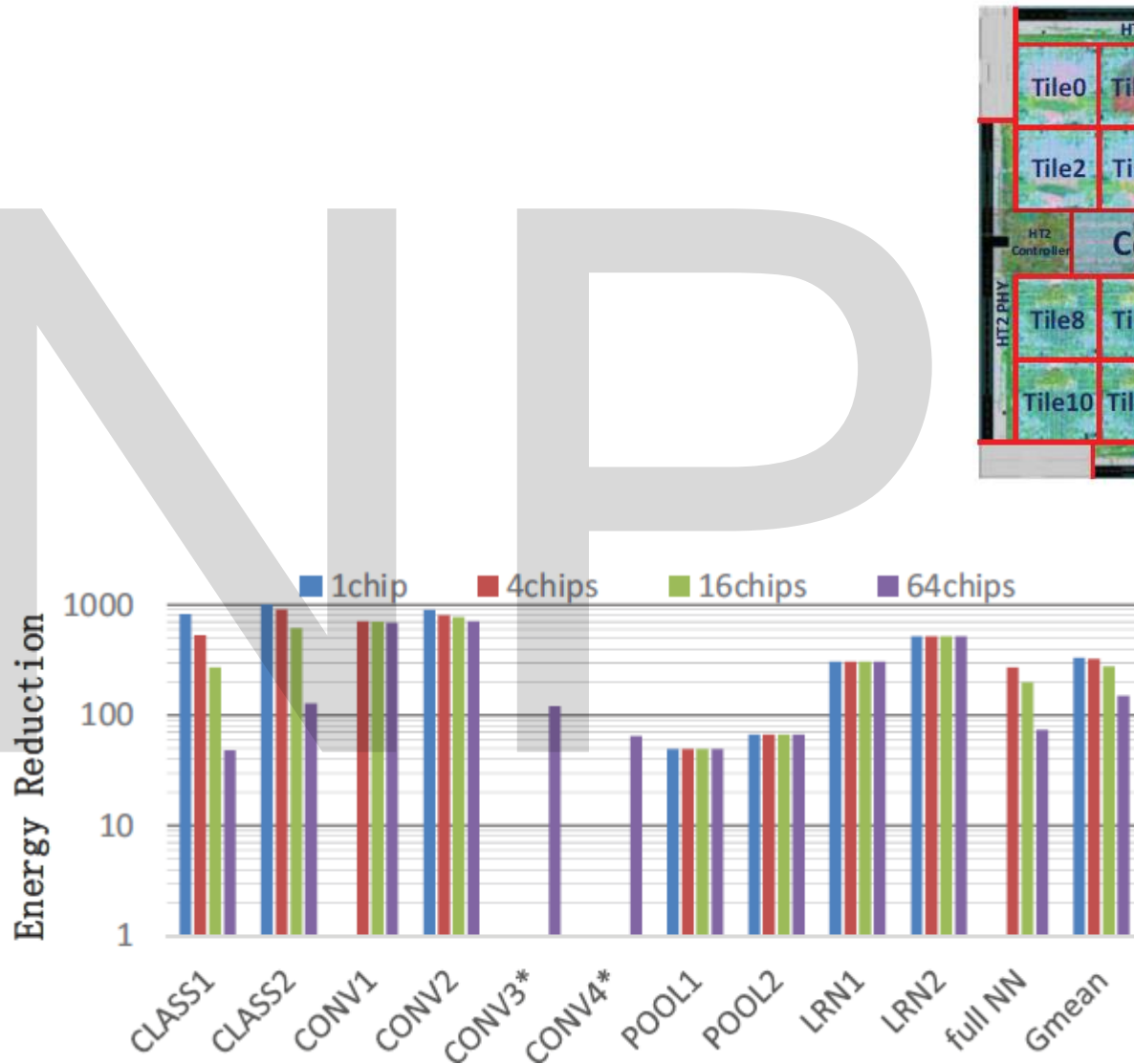


5x5 Crossbar Architecture

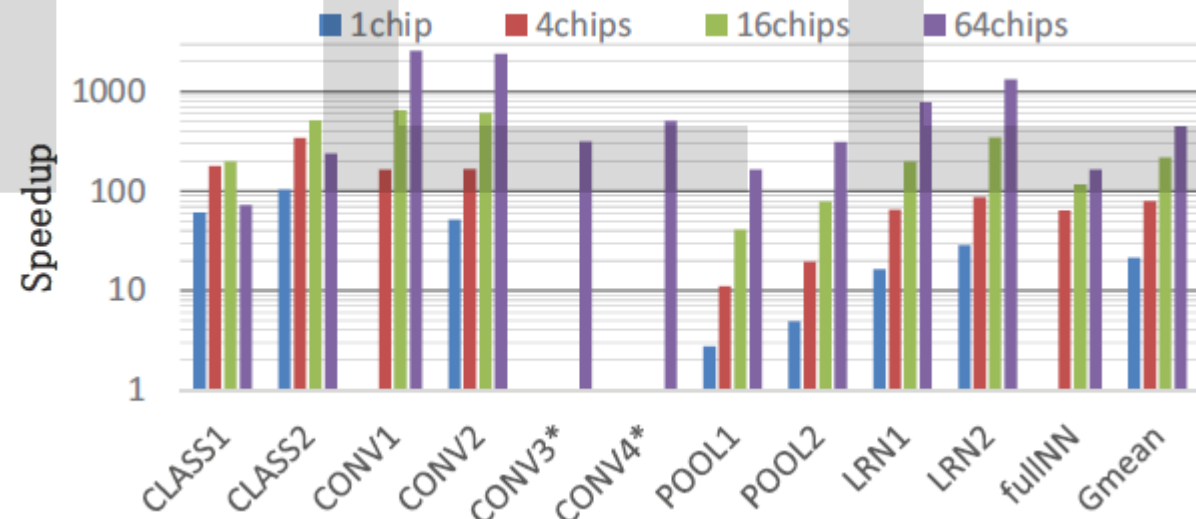
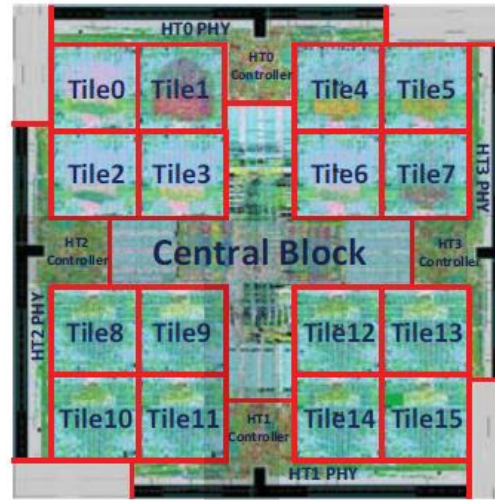
- Five input ports and five output ports
 - North
 - East
 - South
 - West
 - Self
- Consists of multiplexers
- High parallelism
- Reduced head of line blocking



Results



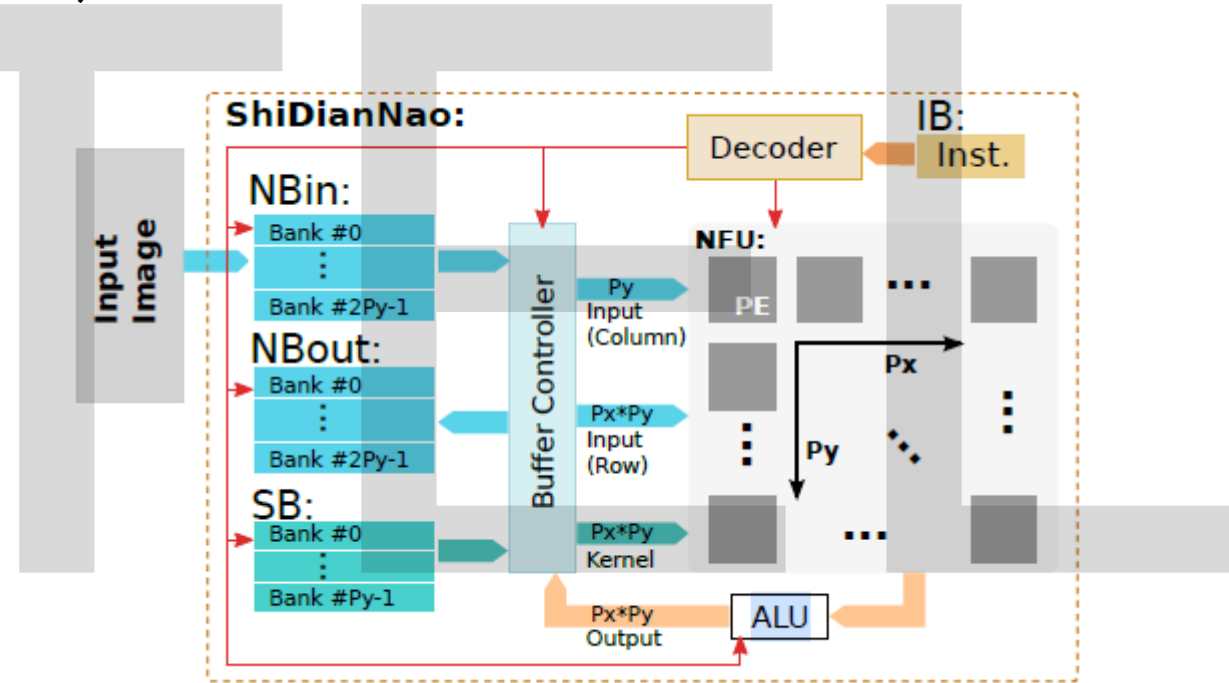
Energy reduction with respect to NVIDIA K20 GPU



Speed up with respect to NVIDIA K20 GPU

ShiDianNao (Vision Computer)

- **Additional components**
 - ALU (to incorporate different activations)
 - Instruction buffer
- **Dataflow**
- **Algorithm-hardware mapping**
- **Executing DNNs**
 - Instead of a single type of layer
- **Application**
 - Facial recognition
 - Handwritten digit recognition





THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)

